The empirical recurrence for OEIS A237804: recovering a conjecture that is not written down

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Abstract

OEIS A237804 records "Empirical recurrence of order 87", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 194$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 87, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 87 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237804 is "Number of (n+1)X(3+1) 0..2 arrays with the maximum plus the upper median minus the minimum of every 2X2 subblock equal.". It has offset 1 and begins

$$1085, 10433, 94325, 932255, 9105033, 91394885, 915038911, 9256755467, \ \dots$$

The entry states:

> Empirical recurrence of order 87 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 10:29:29, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 194$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 194$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 388$ exact terms of the model returns order 87 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{87} c_i\, a(n-i),$$

c_1	18	c_{23}	−19764126150364926830	c_{45}	−20345770213690112028932763	c_{67}	94
c_2	217	c_{24}	14836482474933147940	c_{46}	−29248497595062500439781612	c_{68}	−1
c_3	−5294	c_{25}	13563732063040913356	c_{47}	37045806533320857022536584	c_{69}	−2
c_4	−16403	c_{26}	−130404408691998761896	c_{48}	40026867790058529737492767	c_{70}	4
c_5	704948	c_{27}	−68263929204891372063	c_{49}	−56834491006726781217519707	c_{71}	4
c_6	82705	c_{28}	95545607362896058890	c_{50}	−45851699063594239332346870	c_{72}	−
c_7	−56584136	c_{29}	20172129736113491784	c_{51}	73557528686746167002037132	c_{73}	−
c_8	78865362	c_{30}	−586997618040202206057	c_{52}	43550594932175151132613717	c_{74}	1
c_9	3076026863	c_{31}	25939303316961984234	c_{53}	−80288895649150119915495756	c_{75}	
c_{10}	−7199166181	c_{32}	303765133651797506468	c_{54}	−33778782955352208756603085	c_{76}	−
c_{11}	−120637593501	c_{33}	−781879798397870625689	c_{55}	73791982911443384289980890	c_{77}	
c_{12}	365564762756	c_{34}	−1328612185521813333072	c_{56}	20824592946090046063790339	c_{78}	
c_{13}	3549504142917	c_{35}	583778637790523909641	c_{57}	−56948955132985558274367151	c_{79}	
c_{14}	−12790096388590	c_{36}	4923495275581916654426	c_{58}	−9637602128018719030465208	c_{80}	
c_{15}	−80402546736359	c_{37}	−2946693768458573576900	c_{59}	36759718260662043172253570	c_{81}	
c_{16}	332992110159246	c_{38}	−15481970553464714734063	c_{60}	2815038601185442749958555	c_{82}	
c_{17}	1426499548921077	c_{39}	11478827904633556040017	c_{61}	−19743137517971584389044423	c_{83}	
c_{18}	−6717451929297421	c_{40}	41336566207469305979175	c_{62}	−887584599139955490384	c_{84}	
c_{19}	−20034163844778889	c_{41}	−36128226252506591036475	c_{63}	8765263160568774888691411	c_{85}	
c_{20}	107696548314961391	c_{42}	−93686633263893248108386	c_{64}	−5431397445477791655730	c_{86}	
c_{21}	223653595364293632	c_{43}	93844739100887537912924	c_{65}	−3190541833737192571081290	c_{87}	
c_{22}	−1396521986902156263	c_{44}	180009350513544590390975	c_{66}	35974555820991118596099		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 87, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $87 + 4$ terms removed returns order 87 as well, so $\deg q_{\min} = 87 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ☐

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $87+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 87 that this sequence satisfies — and that family turns out to have exactly one member.

References

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